THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 11: **Keeping Time with the Skies** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The new moon of 1 January 2024 is followed by a full moon after about: 1

(a) 15 days **(b)** 60 days **(c)** 7 days **(d)** 30 days

2. Which of the following is correct about summer solstice in the northern hemisphere? 1

(a) the southern hemisphere has the longest day
(b) the northern hemisphere is tilted towards the Sun
(c) the day and night are equal
(d) the Earth is nearest to the Sun

3. Assertion (A): Seasons are caused by the tilt of the Earth's axis. Reason (R): The Earth is closest to the Sun in summer in India. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. If today is the first quarter, the Moon will be full in about: 1

(a) 14 days **(b)** 21 days **(c)** 1 day **(d)** 7 days

5. During the winter solstice in the northern hemisphere the Sun at noon is: 1

(a) overhead **(b)** lowest in the sky
(c) highest **(d)** below the horizon all day

6. Assertion (A): A solar eclipse occurs on a new moon day. Reason (R): The Moon comes between the Sun and the Earth. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A year has 365 days. How many weeks and extra days does it have? 1

(a) 53 weeks **(b)** 52 weeks and 1 day
(c) 52 weeks **(d)** 51 weeks and 3 days

8. In which month does India's monsoon begin approximately? 1

(a) December **(b)** October **(c)** June **(d)** March

9. Assertion (A): Stars appear to move in circles around the Pole Star. Reason (R): The Earth rotates on its axis. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. If 1 January 2023 was a Sunday, then 1 January 2024 was a: 1

(a) Saturday **(b)** Tuesday **(c)** Monday **(d)** Sunday

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do the Sun and the Moon appear of almost the same size in the sky? 2

12. Why is it difficult to measure noon accurately using a short stick? 2

13. Why does the Moon appear larger near the horizon? Is it really bigger? 2

14. Why do lunar months have either 29 or 30 days in a calendar? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the day of the week 100 days from a Monday. 3

16. Explain how a total solar eclipse is different from an annular eclipse. 3

17. Why does the Moon seem to follow us as we travel? 3

OR

Explain how to find the latitude of Jaipur by observing the Pole Star and describe the instrument. 3

18. Explain how the Moon's shadow causes the solar eclipse and why only a small part of the Earth sees totality. 3

OR

How can the passing of the lunar months be observed in a village without a calendar? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe how the Moon's phases and eclipses are connected to the positions of the Sun, Earth and Moon, and why eclipses do not occur every month. 5

OR

Explain how a sundial is made at a place and why it does not work at night or on cloudy days. State two of its limitations. 5

20. Compare the Indian lunisolar calendar with the Gregorian calendar and explain why Diwali and Eid follow different patterns in the Gregorian calendar. 5

OR

Calculate and discuss: if the Earth rotated twice as fast, what changes would take place in the length of the day, and the number of days in a year? Compare with the Moon. 5

END OF QUESTION PAPER — **Total: 40 Marks**